Pairing instability. Superconductivity.

$$H = H_0 + H'(t) \quad , H'(t) = \lambda(t) \sum_k c^+_{k\uparrow} c^+_{-k\downarrow} + h.c.$$

Then $\langle \sum_k c_{-k\downarrow} c_{k\uparrow} \rangle = \int_{-\infty}^{\infty} dt' \; -i\theta(t-t') \langle [\sum_k c_{-k\downarrow}(t) c_{k\uparrow}(t),$

$$\sum_{k'} c^+_{k'\uparrow} c^+_{-k'\downarrow}] \rangle_{H_0} \lambda(t) = \int_{-\infty}^{t} dt' \; \chi_{pair}(t) \lambda(t).$$

Calculate pair susceptibility : $\hat{c}_{k\sigma}(t) = e^{-i\xi_\sigma(k)t} \hat{c}_{k\sigma}$

$$\chi_{pair}(t) = -i\theta(t) \sum_{kk'} e^{-i\xi_\downarrow(-k)t} e^{-i\xi_\uparrow(k)t} \cdot$$

$$\cdot \langle [\hat{c}_{-k\downarrow} \hat{c}_{k\uparrow} , \hat{c}^+_{k'\uparrow} \hat{c}^+_{-k'\downarrow}] \rangle$$

$$[\;,\;] = \hat{c}_{-k\downarrow} \hat{c}_{k\uparrow} \hat{c}^+_{k'\uparrow} \hat{c}^+_{-k'\downarrow} - \hat{c}^+_{k'\uparrow} \hat{c}^+_{-k'\downarrow} \hat{c}_{-k\downarrow} \hat{c}_{k\uparrow}$$

$$\Downarrow$$

$$\hat{c}_{-k\downarrow} \hat{c}^+_{-k'\downarrow} \hat{c}_{k\uparrow} \hat{c}^+_{k'\uparrow} = (\delta_{kk'} - \hat{c}^+_{-k'\downarrow} \hat{c}_{-k\downarrow})(\delta_{kk'} - \hat{c}^+_{k'\uparrow} \hat{c}_{k\uparrow})$$

$$= \delta_{kk'} - \delta_{kk'} (\hat{c}^+_{k'\uparrow} \hat{c}_{k\uparrow} + \hat{c}^+_{-k'\downarrow} \hat{c}_{-k\downarrow})$$

$$+ \hat{c}^+_{-k'\downarrow} \hat{c}_{-k\downarrow} \hat{c}^+_{k'\uparrow} \hat{c}_{k\uparrow} =$$

$$= \hat{c}^+_{k'\uparrow} \hat{c}^+_{-k'\downarrow} \hat{c}_{-k\downarrow} \hat{c}_{k\uparrow} + \delta_{kk'} (1 - \hat{c}^+_{k\uparrow} \hat{c}_{k\uparrow} - \hat{c}^+_{-k\downarrow} \hat{c}_{-k\downarrow})$$

$$\Rightarrow [\;,\;] = \delta_{kk'} (1 - c^+_{k\uparrow} c_{k\uparrow} - c^+_{-k\downarrow} c_{-k\downarrow}).$$

$$\chi_{pair}(t) = -i\theta(t) \sum_k e^{-i(\xi_\downarrow(-k) + \xi_\uparrow(k))t} \left[1 - f(\xi_\uparrow(k)) - f(\xi_\downarrow(-k))\right]$$

For system with inversion symmetry ($\vec{r} \to -\vec{r}$)

$$\xi_\sigma(-k) = \xi_\sigma(k).$$

For spin-rotationally invariant system

$$\xi_\uparrow = \xi_\downarrow.$$

Thus $\chi_{pair}(t) = -i\,\theta(t) \sum_k e^{-i2\xi(k)t}\left(1-2f(\xi_k)\right)$

$$\chi_{pair}(\omega) = \int dt\,\left(-i\theta(t)\right) e^{i\omega t} \sum_k e^{-i2\xi(k)t}\left(1-2f(\xi_k)\right)$$

$$= -i \int_0^\infty dt\; e^{i(\omega+i\delta)t} \sum_k e^{-i2\xi(k)t}\,\frac{e^{\beta\xi_k}-1}{e^{\beta\xi_k}+1}$$

$$= -i \sum_k \frac{-1}{i(\omega+i\delta-2\xi_k)}\tanh\left(\frac{\beta\xi_k}{2}\right) =$$

$$= \sum_k \frac{\tanh(\beta\xi_k/2)}{\omega+i\delta-2\xi_k}$$

Pair susceptibility per unit volume is

$$\chi_{pair}(\omega) = \int \frac{d\vec{k}}{(2\pi)^3}\,\frac{\tanh(\beta\xi_k/2)}{\omega-2\xi_k+i\delta} \approx \nu\int_{-\Lambda}^{\Lambda} d\xi\,\frac{\tanh\left(\frac{\beta\xi}{2}\right)}{\omega-2\xi+i\delta}$$

where $\Lambda \sim E_F$ is high-energy cutoff.

Static susceptibility $\chi_{pair}(\omega=0) = -\frac{1}{2}\nu\int_{-\Lambda}^{\Lambda} d\xi\,\frac{th(\beta\xi/2)}{\xi}$

$$= \left(x=\frac{\beta\xi}{2}\right) = -\frac{1}{2}\nu\int_{-\beta\Lambda/2}^{\beta\Lambda/2} dx\,\frac{th\,x}{x} = -\nu\int_0^{\beta\Lambda/2} dx\,\frac{th\,x}{x}.$$

$th\,x \to \begin{cases} 1, & x\to\infty \\ x, & x\ll 1\end{cases} \Rightarrow \chi_{pair}$ diverges logarithmically at upper limit

$$\chi_{pair}(\omega=0) = -\nu\ln\left(c\frac{\beta\Lambda}{2}\right) = -\nu\ln\left(\frac{c\Lambda}{2T}\right) \quad (c=\text{const})$$

$\chi_{pair}\to\infty$ as $T\to 0 \Rightarrow$ expect pairing instability at low T.

This is generic result $\Rightarrow$ all conductors must become superconductors at sufficiently low T!

But we need <u>attractive</u> interaction for that to happen.

Interaction $H_{int} = V = \frac{1}{2V} \sum_{\substack{k_1 k_2 q \\ \sigma_1 \sigma_2}} v(q)\, c^+_{k_1+q\sigma_1} c^+_{k_2-q\sigma_2} c_{k_2\sigma_2} c_{k_1\sigma_1}$

Simplify to contact interaction

$$\Rightarrow H_{int} = \frac{U}{V} \sum_{k_1 k_2 q} c^+_{k_1+q\uparrow} c^+_{k_2-q\downarrow} c_{k_2\downarrow} c_{k_1\uparrow} =$$

Select a subset: terms with $\vec{k}_2 = -\vec{k}_1$

$$\Rightarrow H_{int} = \underbrace{\frac{U}{V} \sum_{k_1 q} c^+_{k_1+q\uparrow} c^+_{-k_1-q\downarrow} c_{-k_1\downarrow} c_{k_1\uparrow}}_{H_{BCS}} + \underbrace{\frac{U}{V} \sum_{k_2 \neq -k_1, q} \cdots}_{\text{the rest}}$$

H_{rest} modifies quasi-particles but does not cause superconducting pairing $\Rightarrow$ assume it can be neglected.

$$H_{BCS} = \frac{U}{V} \underbrace{\sum_{k_1} \hat{c}^+_{k_1\uparrow} \hat{c}^+_{-k_1\downarrow}}_{\hat{\Delta}^+} \underbrace{\sum_{k_2} \hat{c}_{-k_2\downarrow} \hat{c}_{k_2\uparrow}}_{\hat{\Delta}}$$

$$\approx \frac{U}{V} \hat{\Delta}^+ \hat{\Delta} = \frac{U}{V} \left(\hat{\Delta}^+ + \bar{\Delta}^+ - \bar{\Delta}^+\right)\left(\hat{\Delta} + \bar{\Delta} - \bar{\Delta}\right)$$

Here $\bar{\Delta} = \langle \hat{\Delta} \rangle = \langle \sum_k c_{-k\downarrow} c_{k\uparrow} \rangle$.

$$H_{BCS} \approx \frac{U}{V} \left(\delta\hat{\Delta}^+ + \bar{\Delta}^+\right)\left(\delta\hat{\Delta} + \bar{\Delta}\right) =$$

$$\cong \frac{U}{V} \left\{ \bar{\Delta}^+ \bar{\Delta} + \left(\hat{\Delta}^+ - \bar{\Delta}^+\right)\bar{\Delta} + \left(\hat{\Delta} - \bar{\Delta}\right)\bar{\Delta}^+ \right\}$$

$$= \frac{U}{V} \left\{ \hat{\Delta}^+ \bar{\Delta} + \hat{\Delta}\bar{\Delta}^+ - \bar{\Delta}^+\bar{\Delta} \right\}$$

operator t-dependent expectation value

$$\Rightarrow H = H_0 + H_{BCS} + H'(t) \longrightarrow H_0 + \Big\{ \lambda(t) \sum_k c^\dagger_{k\uparrow} c^\dagger_{-k\downarrow}$$

$$+ \frac{U}{V} \bar{\Delta}_t \sum_k c^\dagger_{k\uparrow} c^\dagger_{-k\downarrow} + h.c. \Big\} \Rightarrow \lambda(t) \to \lambda(t) + \frac{U}{V} \bar{\Delta}_t$$

Linear response tells us that

$$\underbrace{\Big\langle \sum_k c_{-k\downarrow} c_{k\uparrow} \Big\rangle_\omega}_{\bar{\Delta}_\omega} = \chi^{(0)}_{pair}(\omega) \left(\lambda(\omega) + \frac{U}{V} \bar{\Delta}_\omega \right)$$

$$\Rightarrow \bar{\Delta}_\omega = \frac{\chi^{(0)}_{pair}(\omega) \; \lambda(\omega)}{1 - \frac{U}{V} \chi^{(0)}_{pair}(\omega)} = \chi_{pair}(\omega) \; \lambda(\omega).$$

Instability when $1 - \frac{U}{V} \sum_k \frac{\tanh(\beta \xi_k / 2)}{\omega + i\delta - 2\xi_k} = 0$

In particular, $\omega = 0$ divergence if $1 = U \chi^{(0)}_{pair}(\omega = 0)$

$$\Rightarrow \quad 1 = U \left\{ -\nu \ln\left(\frac{c\Lambda}{2T} \right) \right\}.$$

Possible if $\underline{U < 0}$: attraction!

$$\frac{1}{\nu |U|} = \ln\left(\frac{c\Lambda}{2T_c} \right), \quad T_c = \frac{1}{2} c\Lambda \, e^{-\frac{1}{\nu |U|}}$$

Response of a superconductor to a static external "pair" field $H_{pair} = \sum_{k,q} \left(v_q \, c^+_{k\uparrow} c^+_{-k+q\downarrow} + v_q^* \, c_{-k+q\downarrow} c_{k\uparrow} \right)$

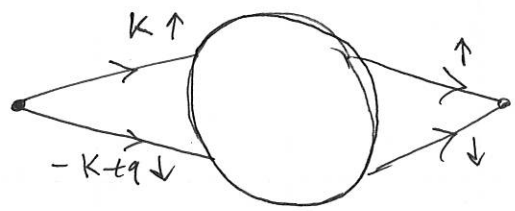
Full Hamiltonian $H = H_0 + V + H_{pair}$.

According to linear response theory

$$\left\langle \sum_k c_{-k+q\downarrow} c_{k\uparrow} \right\rangle_t = \int dt' \left(-i\,\theta(t-t') \right) \cdot$$

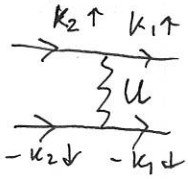
$$\cdot \left\langle \left[\sum_k c_{-k+q\downarrow}(t)\, c_{k\uparrow}(t) \,,\, \sum_{k'} c^+_{k'\uparrow} c^+_{-k'+q\downarrow} \right] \right\rangle_{H_0+V} \cdot v_q$$

Then in Matsubara technique (see pp. 70-71 of Rickayzen) we study $G_{pair}(k, q, \tau) = \mathrm{Tr}\left\{ \hat{\rho}_0 \; c_{-k+q\downarrow}(\tau)\, c_{k\uparrow}(\tau)\, c^+_{k\uparrow}(0)\, c^+_{-k+q\downarrow}(0) \right\}$

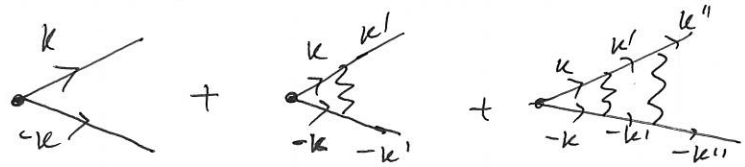


In a free system ($V=0$) there are no interaction lines between single particle GF $G^{(0)}_\uparrow(k,\omega)$ and $G^{(0)}_\downarrow(-k+q,\omega')$.

But if we take $V = H_{BCS} = \frac{U}{V} \sum_{k_1 k_2} c^+_{k_1\uparrow} c^+_{-k_1\downarrow} c_{-k_2\downarrow} c_{k_2\uparrow}$



$\Rightarrow$



$= \Lambda(k,\omega)$

$\Lambda(k,\omega) =$

$$\Lambda(k,\omega) = -U + \left(-\frac{U}{V}\right) \frac{1}{\beta} \sum_{\omega'} \sum_{k'} G(k',\omega')\, G(-k',-\omega')\, \Lambda(k',\omega')$$

For such constant U Λ does not depend on $\vec{k}, \omega$

$$\Rightarrow \Lambda = -U + \frac{1}{\beta}\sum_{\omega'} \frac{-U}{V}\sum_{k'} G(k',\omega')\, G(-k',-\omega')\, \Lambda$$

$$A = \frac{1}{\beta}\sum_{\omega'}\frac{1}{V}\sum_{k'} G(k',\omega')\, G(-k',-\omega') = \frac{1}{\beta}\sum_{\omega'}\int (dk)\, \frac{1}{i\omega_n - \xi_k}\, \frac{1}{-i\omega_n - \xi_{-k}}$$

$$(\xi_k = \xi_{-k}) \quad = \frac{1}{\beta}\sum_{\omega}\underbrace{\int \frac{d\vec{k}}{(2\pi)^3}}_{\nu\int_{-\infty}^{\infty} d\xi}\, \frac{1}{\omega_n^2 + \xi_k^2} = \frac{\nu}{\beta}\sum_{\omega_n}\frac{\pi}{|\omega_n|}$$

$$\omega_n = \pi T(2n+1)$$

$$A = \nu T \pi\, 2\sum_{n=0}^{\infty}\frac{1}{2\pi T\left|n+\frac{1}{2}\right|} = \nu\sum_{n=0}^{\infty}\frac{1}{\left|n+\frac{1}{2}\right|}$$

The sum diverges, needs to be cut off at some n_{max}

i.e. $\omega_{max} = 2\pi T\left(n_{max} + \frac{1}{2}\right)$

$$n_{max} = \frac{\omega_{max}}{2\pi T}$$

$$\sum_{n=0}^{n_{max}}\frac{1}{n+\frac{1}{2}} = \ln\left(\frac{4e^{\gamma}}{\pi}\, n_{max}\right) = \ln\left(\frac{4e^{\gamma}}{\pi}\,\frac{\omega_{max}}{2\pi T}\right) =$$

$$= \ln\left(\frac{2e^{\gamma}}{\pi}\,\frac{\omega_D}{T}\right)$$

where $\omega_{max} \to \omega_D$ is the max frequency to which interaction U is const.

Then $\Lambda = \dfrac{-U}{1+UA} = \dfrac{-U}{1+\nu U \ln\left(\frac{c_1\omega_D}{T}\right)}$

diverges for $U<0$ at $T_c = c_1\omega_D e^{-\frac{1}{\nu U}}$ Superconductivity!!!

(Summary from Bruus-Flensberg, Ch. 3)

Electron-phonon interaction comes from electron-ion interaction $V_{el-ion} = \underbrace{V_{el-latt}}_{} + \underbrace{V_{el-ph}}_{}$

interaction with periodic lattice. Leads to band structure.

interaction with deviations (vibrations) of the ideal lattice. Leads to dynamic scattering.

$$V_{el-ion} = \int d\vec{r}\,(-e)\,\rho_{el}(\vec{r}) \sum_{j=1}^{N} \underbrace{V_{ion}(\vec{r}-\vec{R}_j)}_{V_{ion}(\vec{r}-\vec{R}_j^{\,0}) - \vec{\nabla}_r V_{ion}(\vec{r}-\vec{R}_j^{\,0})\cdot\vec{u}_j}$$

$$\vec{R}_j = \underbrace{\vec{R}_j^{\,0}}_{\text{equilibrium ion position}} + \vec{u}_j$$

fluctuations

$$\Rightarrow V_{el-ph} = \int d\vec{r}\,\rho_{el}(\vec{r}) \left\{ \sum_j e\vec{u}_j \cdot \vec{\nabla}_r V_{ion}(\vec{r}-\vec{R}_j^{\,0}) \right\}$$

electrons couple to gradient of V_{ion} via local displacements $\vec{u}_j$

$$\nabla_r V_{ion}(\vec{r}-\vec{R}_j^{\,0}) = \frac{1}{V} \sum_{\substack{\vec{q}\in BZ \\ \text{(Brill. Zone)}}} \sum_{\substack{\vec{G}\in RL \\ \text{(Recip. Latt.)}}} i(\vec{q}+\vec{G})\, V_{\vec{q}+\vec{G}}\, e^{i(\vec{q}+\vec{G})\cdot(\vec{r}-\vec{R}_j^{\,0})}$$

$$\vec{u}_j = \frac{1}{\sqrt{N}} \sum_{\vec{k}\in BZ} \sum_{\substack{\lambda \\ \text{(polarization)}}} \sqrt{\frac{\hbar}{2M\omega_k}} \left(b_{k,\lambda} + b^{+}_{-k,\lambda}\right) \underbrace{\vec{\varepsilon}_{k\lambda}}_{\text{polarization vector}} e^{i\vec{k}\cdot\vec{R}_j^{(0)}}$$

$$\Rightarrow \sum_j \vec{u}_j \cdot \vec{\nabla}_r V_{ion}(\vec{r}-\vec{R}_j^{\,0}) \sim \sum_j e^{i(\vec{q}+\vec{G})\cdot(\vec{r}-\vec{R}_j^{\,0})}\, e^{i\vec{k}\cdot\vec{R}_j^{\,0}}$$

$$\sim e^{i(\vec{q}+\vec{G})\cdot\vec{r}} \underbrace{\sum_j e^{i\vec{R}_j^{\,0}(\vec{k}-\vec{q}-\vec{G})}}_{\sum_j e^{i\vec{R}_j^{\,0}(\vec{k}-\vec{q})} = N\delta_{\vec{k},\vec{q}}}$$

(because $\vec{R}_j^{\,0}\cdot\vec{G} = 2\pi\cdot\text{Integer}$)

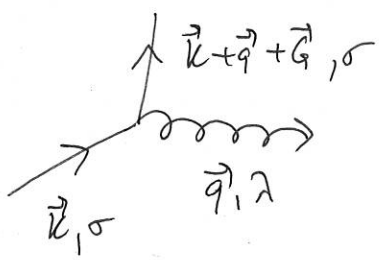
So that $\sum_j e\vec{u}_j \cdot \vec{\nabla}_r V_{ion}(\vec{r}-\vec{R}_j^0) = \frac{1}{V} \sum_{\substack{\vec{q}\in BZ \\ \vec{G}\in RL}} g_{\vec{q},\vec{G},\lambda} \left(b_{\vec{q},\lambda} + b^+_{-\vec{q},\lambda}\right) e^{i(\vec{q}+\vec{G})\cdot\vec{r}}$

where $g_{\vec{q},\vec{G},\lambda} = ie\sqrt{\frac{N\hbar}{2M\omega_{q\lambda}}}\,(\vec{q}+\vec{G})\cdot\vec{\varepsilon}_{q\lambda}\, V_{\vec{q}+\vec{G}}$

is el-phonon coupling constant

Since el. density $\rho_{el}(\vec{r}) = \frac{1}{V}\sum_{\vec{k}\vec{p}\sigma} e^{-i\vec{p}\cdot\vec{r}}\, c^+_{k+p\sigma}\, c_{k\sigma}$

$$\Rightarrow V_{el-ph} = \frac{1}{V}\sum_{\vec{k}\sigma}\sum_{\vec{q}\lambda}\sum_{\vec{G}} g_{\vec{q},\vec{G},\lambda}\, c^+_{\vec{k}+\vec{q}+\vec{G},\sigma}\, c_{\vec{k},\sigma}\left(b_{\vec{q},\lambda} + b^+_{-\vec{q},\lambda}\right)$$



(1) Normal process: $\vec{G}=0$
Umklapp : $\vec{G}\neq 0$
$V_{\vec{q}+\vec{G}} \sim \frac{1}{(\vec{q}+\vec{G})^2}$ so that often Umklapp processes can be neglected.

(2) For isotropic medium $\vec{q}\cdot\vec{\varepsilon}_{q\lambda} \neq 0$ for longitudinal oscillations

$\Rightarrow$ often enough to consider Isotropic Normal Acoustical processes when $g_{\vec{q},\vec{G}=0,\lambda} \sim q$

$$\Rightarrow V_{el-ph} = \frac{1}{V}\sum_{\vec{k}\sigma}\sum_{\vec{q}} g_{\vec{q}}\, c^+_{\vec{k}+\vec{q},\sigma}\, c_{\vec{k},\sigma}\left(b_q + b^+_{-q}\right)$$

For $+eZ$ charged ions interacting with electrons with Yukawa potential $g_{\vec{q}} = i\,4\pi Z e^2 \frac{q}{q^2+K_{scr}^2}\sqrt{\frac{N\hbar}{2M\omega_q}}$

Acoustic phonons: $\omega_q = cq$.

$$D^{(0)}(q,\tau) = -\langle T_\tau \; \hat{A}_q(\tau)\, \hat{A}_{-q}(0)\rangle_0$$

$$\hat{A}_q(\tau) = e^{\tau H_{ph}} (b_q + b^+_{-q})\, e^{-\tau H_{ph}}$$

$$H_{ph} = \sum_q \Omega_q \left(b^+_q b_q + \tfrac{1}{2}\right)$$

$$\frac{d\hat{A}_q(\tau)}{d\tau} = H_{ph}\hat{A}_q - \hat{A}_q H_{ph} = -\Omega_q (b_q - b^+_q)$$

$$\sum_k \Omega_k \left[b^+_k b_k,\, b_q + b^+_{-q}\right] = -\Omega_q b_q + \Omega_{-q} b^+_{-q}$$

$$\underbrace{b^+_k b_k b_q} - b_q b^+_k b_k = -b_k \delta_{kq}$$

$$b^+_k b_q b_k = (\delta_{kq} + b_q b^+_k) b_k$$

$$\underbrace{b^+_k b_k b^+_{-q}} - b^+_{-q} b^+_k b_k = \delta_{k,-q}\, b^+_k$$

$$b^+_k (\delta_{k,-q} + b^+_{-q} b_k)$$

$$\Rightarrow \frac{d b_q(\tau)}{d\tau} = -\Omega_q b_q(\tau) \Rightarrow b_q(\tau) = e^{-\Omega_q \tau} b_q$$

$$\frac{d b^+_{-q}(\tau)}{d\tau} = \Omega_q b^+_{-q}(\tau) \Rightarrow b^+_{-q}(\tau) = e^{\Omega_q \tau} b^+_{-q}$$

$$\Rightarrow D^0(q,\tau>0) = -\langle (e^{-\Omega_q \tau} b_q + e^{\Omega_q \tau} b^+_{-q})(b_{-q} + b^+_q)\rangle$$

$$= (-)\left\{ e^{-\Omega_q \tau} \underbrace{\langle b_q b^+_q\rangle}_{1+n_B(\Omega_q)} + e^{\Omega_q \tau} \underbrace{\langle b^+_{-q} b_{-q}\rangle}_{n_B(\Omega_{-q})} \right\}$$

$$\Rightarrow D^{(0)}(q, iq_n) = \int_0^\beta d\tau\, e^{iq_n\tau} D^{(0)}(q,\tau) = -(1+n_B)\int_0^\beta d\tau\, e^{-\Omega_q\tau + iq_n\tau}$$

$$- n_B \int_0^\beta d\tau\, e^{\Omega_q \tau + iq_n\tau}$$

$$\Rightarrow D^{(0)}(\vec{q}, iq_n) = \frac{-(1+n_B)}{iq_n - \Omega_q}\left(e^{\beta(iq_n - \Omega_q)} - 1\right)$$

$$- \frac{n_B}{iq_n + \Omega_q}\left(e^{\beta(iq_n + \Omega_q)} - 1\right)$$

But $n_B(\Omega) \equiv \dfrac{1}{e^{\beta\Omega} - 1}$, $e^{iq_n\beta} = 1$.

$$\Rightarrow D^{(0)} = -\frac{(1+n_B)}{iq_n - \Omega} e^{-\beta\Omega} \underbrace{\left(1 - e^{\beta\Omega}\right)}_{\frac{-1}{n_B}}$$

$$- \frac{n_B}{iq_n + \Omega} \underbrace{\left(e^{\beta\Omega} - 1\right)}_{\frac{1}{n_B}} =$$

$$= + \frac{1}{iq_n - \Omega}\left(e^{\beta\Omega} - 1 + 1\right) e^{-\beta\Omega}$$

$$- \frac{1}{iq_n + \Omega} = \frac{1}{iq_n - \Omega} - \frac{1}{iq_n + \Omega} = \frac{2\Omega_q}{(iq_n)^2 - \Omega_q^2}$$

$D^{(0)}(\vec{q}, iq_n) = \dfrac{2\Omega_q}{(iq_n)^2 - \Omega_q^2}$ is phonon Matsubara GF.

Retarded phonon GF : $D_R^{(0)}(\vec{q}, \omega) = \dfrac{2\Omega_q}{(\omega + i\delta)^2 - \Omega_q^2}$.

Thus $H_0 = H_{el} + H_{ph} = \sum_{k\sigma} \varepsilon_k c^+_{k\sigma} c_{k\sigma} + \sum_q \Omega_q (b^+_q b_q + \frac{1}{2})$

Thus H_0 determines diagrammatic expansion:

$$G_\sigma(\vec{k},\tau) = -\sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \int_0^\beta d\tau_1 \dots \int_0^\beta d\tau_m \langle T_\tau \hat{W}(\tau_1) \dots \hat{W}(\tau_m) \hat{c}_{k\sigma}(\tau) \hat{c}^+_{k\sigma}(0) \rangle^0_c$$

el. GF

connected only

$$\int_0^\beta d\tau_j \hat{W}(\tau_j) = \frac{1}{V} \int_0^\beta d\tau_j \sum_{\vec{k}\sigma} \sum_{\vec{q}} g_{\vec{q}} \hat{c}^+_{\vec{k}+\vec{q}\,\sigma}(\tau_j) \hat{c}_{\vec{k}\sigma}(\tau_j) \hat{A}_q(\tau_j)$$

el-ph interaction

$$\langle T_\tau \hat{A}_{q_1}(\tau_1) \dots \hat{A}_{q_m}(\tau_m) \hat{c}^+_{k+q_1}(\tau_1) \hat{c}_k(\tau_1) \dots \hat{c}^+_{k+q_m}(\tau_m) \hat{c}_k(\tau_m) \rangle^0_c$$

$$= \langle T_\tau \hat{A}_{q_1}(\tau_1) \dots \hat{A}_{q_m}(\tau_m) \rangle^0_e \langle T_\tau c^+_{k+q_1}(\tau_1) c_k(\tau_1) \dots$$
$$\dots \hat{c}^+_{k+q_m}(\tau_m) c_k(\tau_m) \rangle^0_c$$

Since $\hat{A}_q \simeq b_q + b^+_{-q}$, the only non-zero average occurs when $m = 2n$

$$\Rightarrow g(\vec{q}_i) g(\vec{q}_j) \langle T_\tau \hat{A}^{(\tau_i)}_{q_i} \hat{A}^{(\tau_j)}_{q_j} \rangle = \delta_{\vec{q}_j, -\vec{q}_i} \cdot |g(\vec{q}_i)|^2 \langle T_\tau \hat{A}^{(\tau_i)}_{q_i} \hat{A}^{(\tau_j)}_{-q_i} \rangle = -|g_{\vec{q}_i}|^2 D^0(\vec{q}_i, \tau_i - \tau_j) \delta_{\vec{q}_j, -\vec{q}_i}$$

$\Rightarrow$ (1) $(-1)^m \to (-1)^{2n} (-\ )^n = (-1)^n$

(2) Combinatorial factor $\frac{(2n)!}{n!\,n!}$ due to choosing $(m = 2n)$

$\underline{n}$ momenta $\vec{q}_i$ out of $2n$ momenta

(3) $n!$ to combine n remaining momenta,

and $\frac{1}{2^n}$ factor from symmetrization of all contributions

$\Rightarrow \frac{(-1)^m}{m!} \to \frac{(-1)^n}{2^n n!}$

$$\Rightarrow G_\sigma(\vec{k},\tau) = -\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int d\tau_1 \dots \int d\tau_n \langle T_\tau \, \hat{Q}(\tau_1) \dots \hat{Q}(\tau_n) \, \hat{c}_{k\sigma}(\tau) c^+_{k\sigma}(0) \rangle$$

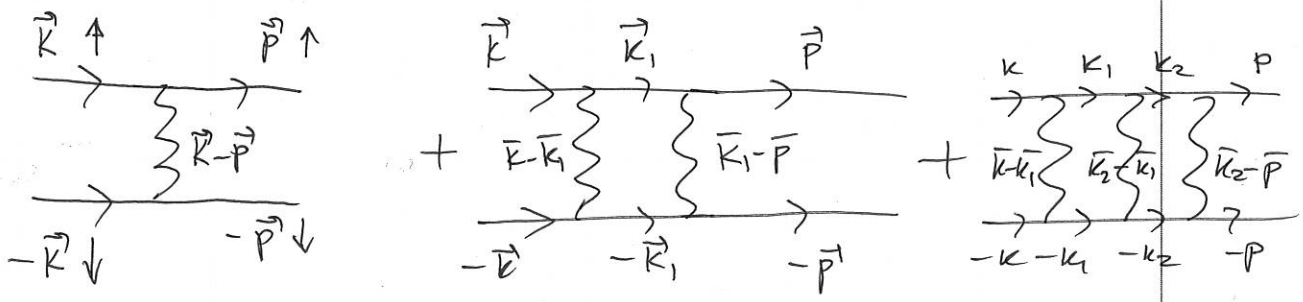
where $\int_0^\beta d\tau_j \, \hat{Q}(\tau_j) = \int d\tau_j \int d\tau_i \sum_{k_1\sigma_1} \sum_{k_2\sigma_2} \sum_q \frac{1}{2V^2} |g_{\vec{q}}|^2 \cdot$

$$\cdot D^0(\vec{q}, \tau_i - \tau_j) \, \hat{c}^+_{k_1+q_1\sigma_1}(\tau_j) \, \hat{c}_{k_2-q_2\sigma_2}(\tau_i) \, \hat{c}_{k_2\sigma_2}(\tau_i) \, \hat{c}_{k_1\sigma_1}(\tau_j)$$

$\Rightarrow$ we can identify a phonon-mediated el-el interaction

$$V^{ph}_{el-el} = \frac{1}{2V} \sum_{k_1\sigma_1} \sum_{k_2\sigma_2} \sum_q \frac{1}{V} |g_{\vec{q}}|^2 D^0(\vec{q}, \tau_i - \tau_j) \, c^+_{k_1+q} \, c^+_{k_2-q} \, c_{k_2} \, c_{k_1}$$

But note that this interaction is dynamic $\rightarrow$ depends on $\tau_i - \tau_j$ $\Rightarrow$ depends on phonon frequency



$$\sim \vec{q}, \omega \Rightarrow D^0_R(\vec{q}, \omega) = \frac{2\Omega_q}{(\omega + i\delta)^2 - \Omega_q^2}$$

$\Rightarrow$ for $\omega = \xi_{\vec{k}} - \xi_{\vec{p}}$ where both ξ are on the fermi-surface

we can approximate $\omega \sim 0$

$$\Rightarrow D^0_R(\vec{q}, 0) \sim \frac{-1}{\Omega_q}$$

$\Rightarrow$ <u>attraction</u> $|g_q|^2 D^0(\vec{q}, \omega) = \begin{cases} -V & \text{for } \omega < \omega_D \\ 0 & \text{for } \omega > \omega_D \end{cases}$